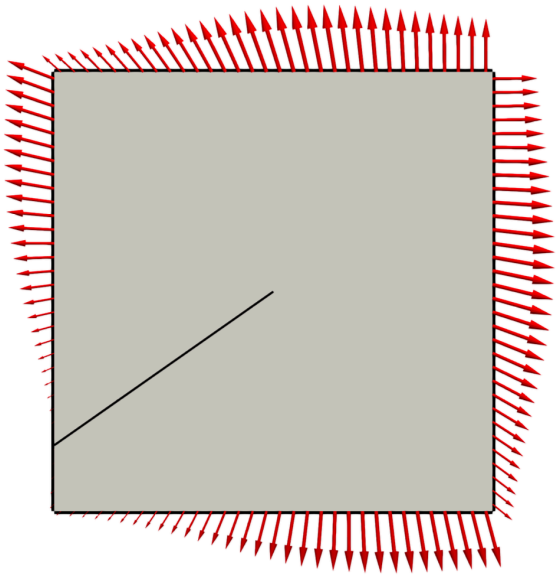
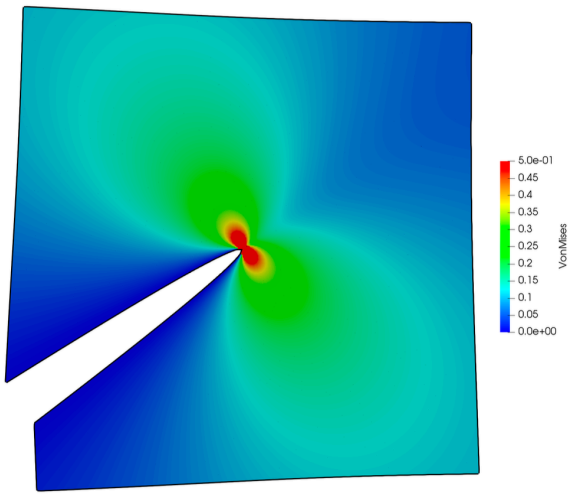
2D Edge Crack Solved with pFEM-GFEM [Sanchez-Rivadeneira & Duarte, 2019]



(a) Domain, crack definition, and applied boundary tractions



(b) Mode I solution and von Mises stress

Inclined Edge Crack, Section 6.4 of [Sanchez-Rivadeneira & Duarte, 2019], available on Canvas: 2019_Quadratic_SGFEM_2D_cmame_published.pdf

1) Go to folder

2d_cracks/2d_edge_crack/pfem-gfem

If you are unable to run ISET on your computer, go to the folder *output_files* and skip to Step 6).

2) Create a folder named 2026_CEE598_Test_1 and copy these files into it

pfem_33x33.tcl
square_2d_domain_tria_33x33.grf

Go to this folder in a terminal window.

3) Check the definition of the crack

a) Open the file *square_2d_domain_tria_33x33.grf* in a text editor.

Modify, if needed, section "BEGIN NODES NUMBERED" of file
square_2d_domain_tria_33x33.grf

Used for horizontal crack, Section 4.1 of paper
100000001 9 8

Used for inclined crack, Section 6.4 of paper
100000001 9 8.6982233047

b) Open the file *pfem_33x33.tcl* in a text editor.

Modify, if needed, the file *pfem_33x33.tcl*

create a crack manager:
#
Inclined edge crack problem, Section 6.4 of paper.
Must also set in section "BEGIN NODES NUMBERED" of *square_2d_domain_tria_33x33.grf*
100000001 9 8.6982233047

crackMgr2D crackMgrID 1 create straightCrackEndPoints tail 0.0 2.414213562 \
tip 8.0 8.0

4) Solve the problem using iset-docker. Run this command in the terminal window

iset-docker pfem_33x33.tcl > pfem_33x33_out.txt

6) Use Paraview (http://www.paraview.org) to open (you can just drop it on ParaView)

pfem_33x33.vtu

a) Show integration elements
b) Visualize the Solution field, component Y.
c) Identify nodes snapped to the crack line (check vertex nodes near the crack tail).

7) Plot the deformed configuration; Zoom in at the crack tip and explain the apparent spurious discontinuity.

8) Plot von Mises stress:

Re-scale
Set Range to max = 0.3

9) Visualize the element-wise error in the energy norm: Field

TrueErrorElemEnergyNorm
TrueRelErrorElemEnergyNorm

10) Visualize field Solution, Y-component, again: Keep this ON as we visualize enrichments

11) Visualize the boundary conditions: Drop *gfem_33x33_BCs.vtu* into ParaView

12) Use ParaView to open *gfem_33x33_GFEM.vtu* (you can just drop it on ParaView)

pfem_33x33_GFEM.vtu

Visualize nodal enrichment (keep Solution field Y on, not the GFEM mesh)

a) BranchFnEnrich : Use red spheres with Radius = 0.5, Scale factor ~= 0.3

b) StepFnEnrich : Use blue spheres with Radius = 0.5, Scale factor ~= 0.2

c) BranchStepPairEnrich : Use yellow spheres with Radius = 0.5, Scale factor ~= 0.2

d) Turn off all GFEM enrichment glyphs, and show "ApproxOrder": Use sphere with Scale Array: " No scale array", Scale factor = 0.1, Coloring: ApproxOrder

Check where each enrichment is used: Element vertices and edges.

Optional: Save ParaView State (.pvsm)

13) If you got lost in the previous Task:

Close ParaView and open it again.

Load ParaView State (you can just drop it on ParaView):

pfem-gfem_enrichments_state_2026.pvsm

Select "Search files under specified directory"

Visualize enrichments, etc.

14) Open the file pfem_33x33_out.txt using a text editor

a) Check the number of branch and step functions created:

CrackManager2D::createBranchFunctions

Number of CompNods enriched with Branch Fn (Including those with BranchStep Pair) = **52**

CrackManager2D::createBranchFunctions

Number of CompNods enriched with BranchStepPair Fn = **4**

Compare the last one with what we saw in ParaView

CrackManager2D::createCrackStepFunctions

Number of CompNods enriched with Step Fn = **77**

Check the problem size:

Num Equations = 9348 Num Non Zeros = 122182

Check the number of positive eigenvalues

MumpsSolver<T>::factorize():

Number of negative pivots = 0
Number of positive eigenvalues = 9348
Dimension of matrix = 9348

Check the convergence of Babuska's algorithm

SymSolver<T>::solve: Stabilization being used:

Iteration: 0 rel_err =
[2.155196802334206e-24]
rel_err_norm = 2.155196802334206e-24 tolerance mrhs = 1e-12

Strain Energy and Error in the Energy Norm

energy:
[3.792933697716725]
energy_exact_sol:
[3.793036132949721]

||u - u_pl||_E = 1.4313296824726039e-02

||u - u_pl||_E / ||u||_E = 5.1967424033667322e-03

15) **Open the Tcl File**

Study the Tcl commands.

Details are in iset_reference_manual.pdf, which is available on Canvas.

OPTIONAL:

Make changes to the Tcl file,

Run the code again and reload files in ParaView

Check the changes in the data printed to the screen after modifications, such as problem size and the error of the pFEM-GFEM solution.

1) **Turn off**

parSet useBranchHeavisideFnPairWithGeomEnrich off

2) **Turn off geometrical enrichment (i.e., use Topological enrichment)**

crackMgr2D crackMgrID 1 branchFnGeometricEnrich inCylinder radius 2

Change the crack orientation: Move the crack tail to the same as in Section 4 of [Sanchez-Rivadeneira & Duarte, 2019]: Edge crack \defined by [(0.8), (8,8)]

Repeat the tasks for this crack configuration.

Conditioning Control Strategies

1) Solve the problem with the pFEM-GFEM

parSet ShiftedSpBasisEnrichments off

and check the value of the strain energy

energy:
[3.792933697716836]
energy_exact_sol:
[3.793036132949722]

2) Solve the problem using the Discontinuous-Shifted pFEM-GFEM

parSet ShiftedSpBasisEnrichments on

and check the value of the strain energy

energy:
[3.792933697716504]

energy_exact_sol:
[3.793036132949721]

3) Solve the problem using the Discontinuous-Interpolant Stable pFEM-GFEM

parSet SGFEMBranchFnEnrichmentsDiscInterp on
parSet ShiftedNotBranchFnEnrichments on

Explain the above commands based on class notes 11a.

check the value of the strain energy

energy:
[3.792933697716674]
energy_exact_sol:
[3.793036132949721]

What conclusions can you draw from the results above?